

Model	Method	Cancer-Myth		CREPE	
		TPR ↑	FPR ↓	TPR ↑	FPR ↓
Qwen2.5 7B	Direct gate	7.0 ^H	10.7 ^H	—	—
	Review CoT gate (2-step)	29.5 ^H	16.1 ^H	—	—
	GEPA	—	—	—	—
	GEPA + text features	—	—	—	—
	Ours (GEPA + SAE)	—	—	—	—
GPT-6 Luna	Direct gate	—	—	—	—
	Review CoT gate (2-step)	—	—	—	—
	GEPA	—	—	—	—
	GEPA + text features	—	—	—	—
	Ours (GEPA + SAE)	—	—	—	—
Gemma 4 12B	Direct gate	40.5 ^H	16.1 ^H	—	—
	Review CoT gate (2-step)	51.3 ^H	32.9 ^H	—	—
	GEPA	—	—	—	—
	GEPA + text features	—	—	—	—
	Ours (GEPA + SAE)	—	—	—	—
Qwen3.8 27B OFF	Direct gate	26.1 ^H	16.1 ^H	—	—
	Review CoT gate (2-step)	57.5 ^H	35.6 ^H	—	—
	GEPA	—	—	—	—
	GEPA + text features	—	—	—	—
	Ours (GEPA + SAE)	—	—	—	—
Qwen3.8 27B ON	Direct gate	44.1 ^H	21.5 ^H	—	—
	Review CoT gate (2-step)	61.4 ^H	38.3 ^H	—	—
	GEPA	—	—	—	—
	GEPA + text features	—	—	—	—
	Ours (GEPA + SAE)	—	—	—	—

Table 1. False-premise detection across models and datasets

Stored results; — means unavailable or not comparable. Values are percentages, one run.

TPR ↑: FPQ detection; FPR ↓: false positives on normal questions.

H: historical full-corpus evaluation (583 FPQ / 149 NFP); valid denominators in ledger.

Model versions, counts, judge/protocol differences and source hashes: result_sources.md.